Notes: Halpern, Koren, and Szeidl (AER, 2015)

Imported Inputs and Productivity

Contents

1 Big picture

2 Data and measurement (key objects)

2.1 Why Hungary is a useful laboratory . . .

2.2 Main data sources and sample construction . . .

2.3 Processing trade and foreign ownership . . .

2.4 Key outcome and input measures . . .

2.5 Descriptive facts . . .

3 Models and empirical specifications (all equations + interpretation)

3.1 Production technology and imported-input mechanism . . .

3.2 Imported varieties, import shares, and firm productivity . . .

3.3 Import choice and fixed costs . . .

3.4 Estimation strategy . . .

3.5 Recovering the deep parameters A and θ . . .

3.6 Heterogeneity across firm types and time . . .

3.7 Fixed-cost estimation . . .

3.8 Aggregate productivity and counterfactuals . . .

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the import effect . . .

4.2 Why the import-share equation is powerful . . .

4.3 How the paper handles import endogeneity . . .

4.4 Robustness to sunk-cost alternatives . . .

4.5 Alternative explanations involving exports, local demand, and ownership . . .

4.6 Foreign ownership: treatment or selection? . . .

4.7 What the paper can and cannot distinguish . . . .
4.8 Relation to the literature . . . .

5 Tables and figures

5.1 Table 1: importers are larger, richer, and more internationally engaged . . . .
5.2 Table 2: the extensive margin is quantitatively important . . . .
5.3 Figure 1 and Table 3: the baseline import gain is large . . . .
5.4 Figure 2 and Table 4: foreign firms get more out of imports . . . .
5.5 Figure 3 and Table 5: acquisition and time variation . . . .
5.6 Table 7 and Figure 4: foreign firms face lower import fixed costs . . . .
5.7 Table 8: imports explain an important part of aggregate productivity growth . . . .
5.8 Table 9 and Figure 5: tariff cuts interact with foreign presence and fixed costs . . . .

6 Short summary

7 Conclusion

7.1 Core conclusion (what the paper establishes) . . . .
7.2 Economic intuition . . . .
7.3 Takeaway . . . .

1 Big picture

- **Question.** How much do imported inputs raise firm productivity, through which mechanism, and for which firms are the gains largest?
- **Setting.** The paper studies Hungarian manufacturing firms from 1992 to 2003 using a merged panel of customs records and firm balance-sheet data. The data observe imported inputs at the product level for essentially the full population of manufacturing firms.
- **Core challenge.** Firms choose imports endogenously. More productive firms may both import more and produce more, so a raw correlation between importing and productivity is not causal.
- **Main conceptual contribution.** The paper estimates a structural model in which imported inputs raise productivity through two distinct channels:
  - **quality**: foreign inputs may have higher price-adjusted quality;
  - **imperfect substitution**: combining domestic and foreign varieties may itself be productive.
- **Economic mechanism.** Firms pay a fixed cost for each imported input variety. More productive firms or firms with lower fixed costs import more varieties. The number of imported products therefore maps into productivity through the model.
- **Main empirical design.** The paper estimates an import-augmented production function using an Olley–Pakes style control-function strategy, De Loecker-style demand controls, and a Gandhi–Navarro–Rivers materials-share restriction.
- **Main baseline result.** In the baseline specification, importing all tradable input varieties raises **revenue productivity by about 22%** and **quantity productivity by about 24–25%**.
- **Mechanism result.** Roughly one-half of the productivity gain comes from **imperfect substitution** between foreign and domestic inputs, and the other half comes from higher **price-adjusted quality** of imports.
- **Foreign-ownership result.** Firms that have been foreign owned use imports more effectively and face lower fixed import costs. The per-dollar productivity gain from imports is substantially larger for foreign firms.
- **Aggregate result.** Between 1993 and 2002, aggregate manufacturing productivity in Hungary rose by **21.1%**. Of this, **5.9 percentage points**—more than one-quarter—are attributed to import-related channels.
- **Policy result.** A tariff cut raises productivity more when the economy already has many importers, when fixed costs of importing are low, and when foreign-owned firms are more prevalent.
- **Economic takeaway.** Imports matter not only because they are cheaper or better; they also matter because foreign and domestic inputs work better together. This makes imported-input access a central margin for firm productivity and aggregate growth.

2 Data and measurement (key objects)

2.1 Why Hungary is a useful laboratory

- During the sample period, Hungarian households and firms used direct securities and customs-based trade records that can be linked at the firm level.
- For manufacturing firms, imported goods are observed at a very fine product level.
- Foreign ownership became quantitatively important in Hungarian manufacturing during the 1990s, which makes it possible to study how foreign-owned and domestic firms differ in their use of imports.
- Trade liberalization, restructuring, and strong growth in imported varieties all happened during the sample window.

Interpretation. Hungary offers unusually rich microdata and enough cross-firm heterogeneity to connect imported inputs, foreign ownership, and productivity in one unified framework.

2.2 Main data sources and sample construction

- Balance-sheet and profit-and-loss data come from the Hungarian Tax Authority for 1992–1999 and from the Hungarian Statistical Office for 2000–2003.
- Trade data come from Hungarian Customs Statistics and report annual exports and imports by firm and by product.
- Products are originally observed at the six-digit Harmonized System level and then aggregated to **HS4** to reduce classification noise.
- The analysis uses imported products classified as intermediate goods, industrial supplies, or capital-good parts in the Broad Economic Categories classification.
- Firms are classified as manufacturing if manufacturing is their primary activity for at least one-half of their lifetime in the data.

The paper constructs two main samples:

- a **main sample**, which excludes firm-year observations classified as processing trade;
- a **firm-level sample**, which excludes firms that are processers for more than half of the years they appear.

After exclusions, the main sample contains **127,472 firm-year observations**.

Interpretation. The main sample is designed for estimation; the firm-level sample is designed to describe aggregate trends without changes in sample composition driven by processing activity.

2.3 Processing trade and foreign ownership

- **Processing trade** is important because customs records can include imported inputs that belong to a foreign party and are re-exported after processing, even though those goods never appear on the firm's balance sheet.

- The paper measures processing trade as the positive part of the difference between customs exports and balance-sheet exports.
- A firm-year is classified as a **processer** if processing trade exceeds **2.5%** of balance-sheet sales.
- About **9%** of observations are classified as processers.

Foreign ownership is defined as follows:

$$\text{Foreign-owned at } t = 1 \quad \text{if the firm is majority foreign-owned at } t \text{ or had been so in the past.} \tag{1}$$

Interpretation. The "has been foreign owned" definition captures the idea that foreign ownership may leave lasting effects on a firm's organization, sourcing, and know-how.

2.4 Key outcome and input measures

The key import-side variables are:

- n_{jt}: the number of imported input products of firm j in year t;
- M^F_{jt}/M_{jt}: the share of imported inputs in total materials expenditure;
- imported products are counted at the HS4 level.

The core production-side variables are:

- output or sales,
- capital,
- labor,
- total materials,
- input and output price indices by industry.

The paper also constructs:

- a county identifier for local demand controls,
- two-digit ISIC industry codes,
- firm export status,
- and an indicator for whether the firm has been foreign owned.

Interpretation. The paper needs both product-level import information and firm-level production data because the number of varieties imported is the key bridge between the structural model and observed productivity.

2.5 Descriptive facts

The paper emphasizes three facts.

Fact 1: Importing is highly heterogeneous.

- About one-half of firms do not import at all.
- Importers are much larger than nonimporters.
- Importers are much more likely to export and to have been foreign owned.
- In Table 1, importers have mean employment of 98.9 workers versus 17.4 for nonimporters, and mean sales of \$6.77 million versus \$0.42 million.
- Conditional on size, firms that have been foreign owned import about **187%** more products than purely domestic firms.

Fact 2: Import spending is concentrated in a few products.

- Among firms importing at least five products, the highest-ranked imported product accounts for about **54%** of total import spending.
- The fifth-ranked product accounts for only about **3.4%**.

Fact 3: The extensive margin is central.

- Average annual import growth is **20.8%**.
- The within-firm intensive margin contributes **17.1** percentage points.
- Positive extensive margins contribute **10.8** percentage points, and negative extensive margins contribute −7.1 percentage points.
- The largest positive extensive-margin component is the addition of **new imported products**, contributing **5.9** percentage points on average.
- Over 1992–2003, total imports rise by about **638%**, with **new firms** contributing **513.4** percentage points.

Interpretation. These facts motivate a model where firms choose *how many* products to import, face product-level fixed costs, and differ strongly in both size and ownership.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production technology and imported-input mechanism

Firm j's output is

$$Q_j = \Omega_j K_j^{\alpha} L_j^{\beta} \prod_{i=1}^{N} X_{ji}^{\gamma_i}, \tag{2}$$

where K_j is capital, L_j is labor, X_{ji} is the composite intermediate input i, and Ω_j is Hicks-neutral productivity.

Each intermediate input combines a domestic and a foreign variety:

$$X_{ji} = \left[(B_{ji} X_{ji}^{F})^{\frac{\theta-1}{\theta}} + (X_{ji}^{H})^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}}, \tag{3}$$

where:

- X_{ji}^F is the foreign input,
- X_{ji}^H is the domestic input,
- B_{ji} captures the quality advantage of foreign inputs,
- θ is the elasticity of substitution between foreign and domestic goods.

The price-adjusted quality advantage of imported inputs is

$$A_{ji} = B_{ji} \frac{P_i^H}{P_i^F}. \tag{4}$$

The effective price of the composite input when the firm imports is

$$P_{ji} = P_i^H \left[1 + A^{\theta-1}\right]^{1/(1-\theta)}. \tag{5}$$

The **per-product import gain** is

$$a = \frac{\log(1 + A^{\theta-1})}{\theta - 1}, \tag{6}$$

and the foreign expenditure share within a composite input is

$$S = \frac{A^{\theta-1}}{1 + A^{\theta-1}}. \tag{7}$$

Interpretation. Imported inputs raise productivity both because A may exceed one and because finite θ makes domestic and foreign inputs complementary rather than perfect substitutes.

3.2 Imported varieties, import shares, and firm productivity

The production importance of the first n_j imported goods is summarized by

$$G(n_j) = \frac{\sum_{i=1}^{n_j} \gamma_i}{\gamma}, \qquad \gamma = \sum_{i=1}^{N} \gamma_i. \tag{8}$$

This gives the import-share equation

$$\frac{M_j^F}{M_j} = S\, G(n_j), \tag{9}$$

where M_j^F is spending on imported inputs and M_j is total intermediate spending.
The log production function becomes

$$q_j = \alpha k_j + \beta l_j + \gamma(m_j - \rho) + a\gamma G(n_j) + \omega_j, \tag{10}$$

where:

- m_j is log material spending,
- ρ is the log price index of domestic composite inputs,
- ω_j is firm productivity.

The revenue production function is

$$r_j - p = \frac{1}{\eta}q + \frac{1}{\eta}v_j + \alpha^* k_j + \beta^* l_j + \gamma^*(m_j - \rho) + \gamma^* a G(n_j) + \omega_j^*, \tag{11}$$

where v_j is a demand shifter, q is industry quantity, and starred coefficients are scaled by $(\eta - 1)/\eta$.

Interpretation. The term $a\gamma G(n_j)$ is the direct productivity contribution of importing more varieties. That is the key object the paper wants to estimate.

3.3 Import choice and fixed costs

Firms pay a fixed cost for each imported product. If the nth imported input has fixed cost f_n, optimal import choice satisfies the marginal-profit inequalities

$$f_{n_j} \leq \pi(n_j) - \pi(n_j - 1), \qquad f_{n_j+1} > \pi(n_j + 1) - \pi(n_j), \tag{12}$$

where $\pi(n)$ denotes expected operating profits when importing n varieties.

Interpretation. More productive firms, or firms with lower fixed import costs, optimally choose more imported varieties. This is the mechanism linking observed import variety to firm characteristics.

3.4 Estimation strategy

The paper estimates the model in three steps.

Step 1: Estimate the shape of $G(n)$ and the import share parameter S.

The paper assumes the parametric form

$$G(n) = \left[1 - \left(1 - \frac{n}{\bar{n}}\right)^{\lambda}\right]\bar{G} \quad \text{for } n \leq \bar{n}, \qquad G(n) = \bar{G} \quad \text{for } n > \bar{n}, \tag{13}$$

where $\lambda \in (0, 1)$ controls curvature and $\bar{G}$ is the tradable-input share.

This is estimated from

$$\frac{M_{jt}^F}{M_{jt}} = S \cdot G(n_{jt}) + u_{jt}. \tag{14}$$

Step 2: Estimate the materials coefficient from the materials first-order condition.

The model implies

$$\gamma^* \frac{\mathbb{E}_{\varepsilon}(R_{jt})}{M_{jt}} = 1. \tag{15}$$

Step 3: Estimate the revenue production function using an Olley–Pakes control function.

Investment is assumed to be monotone in observed productivity:

$$I_{jt} = \xi(\omega_{jt}^{obs}, k_{jt}, l_{jt}, z_{jt}), \tag{16}$$

so it can be inverted to obtain a control function for productivity. The estimating equation is

$$r_{jt} - p_t^s - \gamma^*(m_{jt} - \rho_t^s) = \tilde{h}(I_{jt}, k_{jt}, l_{jt}, z_{jt}) + \delta^* G(n_{jt}) + \varepsilon_{jt}^*, \tag{17}$$

where $\delta^* = \gamma^* a$.

Interpretation. The control function removes the endogenous component of productivity, the materials equation pins down γ^*, and the import-share equation pins down the shape of $G(n)$. Together they identify the productivity gain from importing.

3.5 Recovering the deep parameters A and θ

After estimating a and S, the paper backs out the underlying quality and substitution parameters from

$$\theta = 1 - \frac{\log(1-S)}{a}, \tag{18}$$

$$\log A = a\left[1 - \frac{\log S}{\log(1-S)}\right]. \tag{19}$$

Interpretation. A large productivity gain a combined with only a modest import share S can only be rationalized if domestic and foreign inputs are imperfect substitutes, that is, if θ is not too large.

3.6 Heterogeneity across firm types and time

The paper lets the import-efficiency parameter vary across groups g. In practice, the most important group distinctions are:

- domestic versus foreign-owned firms;
- time periods such as 1992–1994, 1995–1997, 1998–2000, and 2001–2003.

This yields group-specific A_g, S_g, and a_g, while keeping the production technology common across firms.

Interpretation. This is how the paper tests whether foreign-owned firms really get more out of imported inputs and whether import gains changed over the transition period.

3.7 Fixed-cost estimation

The fixed cost of importing product n for firm j in year t is modeled as

$$f_{njt} = \exp\left(\kappa_n^D \, 1_{jt}^D + \kappa_n^F \, 1_{jt}^F + \vartheta_{jt}\right), \tag{20}$$

where 1_{jt}^F indicates that the firm has been foreign owned and ϑ_{jt} is an iid disturbance.

Given the observed n_{jt} and the model-implied profitability of importing additional varieties, the fixed cost schedule is estimated with an ordered probit.

Interpretation. The fixed costs are not directly observed. They are inferred from how aggressively firms expand the number of imported products for a given expected gain from importing.

3.8 Aggregate productivity and counterfactuals

Aggregate productivity is measured as the sales-weighted average of firm log productivity:

$$\Phi_t^R = \sum_j \sigma_{jt}\phi_{jt}^R, \tag{21}$$

where σ_{jt} is firm j's sales share.

The paper then uses the estimated model in a **static partial-equilibrium** environment to:

- decompose aggregate productivity growth in Hungarian manufacturing;
- simulate tariff reductions;
- vary the share of foreign-owned firms;
- vary the fixed costs of importing;
- study the response of domestic input demand under different substitution mechanisms.

Interpretation. The counterfactuals take the estimated firm-level mechanism seriously and ask how much aggregate productivity moves when trade frictions or foreign presence change.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the import effect

There is **no conventional external instrument**. Identification is structural.

The key identifying comparison is between firms that:

- have the same productivity once controlled for,
- face the same demand conditions once controlled for,
- but import different numbers of varieties because they face different fixed costs of importing.

Interpretation. The import gain is identified from residual within-industry, within-demand, within-productivity variation in the number of imported varieties.

4.2 Why the import-share equation is powerful

A major concern would be that more productive firms both import more and spend more on imported inputs. The paper's import-share equation avoids this problem because productivity drops out of the ratio:

$$\frac{M_{jt}^F}{M_{jt}} = S \cdot G(n_{jt}). \tag{22}$$

Interpretation. Productivity may raise both imported-input spending and total materials spending, but it does not mechanically create the cross-firm shape linking the import share to the number of imported products.

4.3 How the paper handles import endogeneity

The paper combines three ideas:

- **Olley–Pakes control function**: use investment to proxy for observed productivity;
- **De Loecker demand controls**: separate demand shocks from productivity;
- **Gandhi–Navarro–Rivers materials restriction**: pin down the materials coefficient without assuming materials are exogenous.

State variables include:

- capital and labor,
- the firm-level demand shifter,
- industry demand,
- industry, year, and county effects,
- foreign-ownership status,
- and, in robustness exercises, past import status and past imported varieties.

Interpretation. The design does not pretend imports are random. Instead, it explicitly models the firm's dynamic decision problem and then controls for the state variables that drive both imports and productivity.

4.4 Robustness to sunk-cost alternatives

One possible critique is that importing may involve sunk costs rather than per-period fixed costs. The paper addresses this in two ways:

- by adding an indicator for being a **previous importer**;
- by adding lagged $\gamma G(n_{j,t-1})$ as a state variable.

The estimated import gain remains positive and economically meaningful:

- $a = 0.194$ when controlling for past import status,
- $a = 0.164$ when controlling for lagged $G(n)$.

Interpretation. Even if importing today partly reflects sunk investments made in the past, imported inputs still deliver large productivity gains.

4.5 Alternative explanations involving exports, local demand, and ownership

The paper controls for exporter status and local demand growth. It also allows foreign-owned and domestic firms to differ in the efficiency of import use.

The main result survives:

- conditioning on exporter status,
- conditioning on past importing,
- conditioning on lagged imported-input variety,
- splitting firms by foreign ownership,
- and allowing import efficiency to vary over time.

Interpretation. The baseline effect is not simply an "international engagement" effect or a short-run proxy for export participation.

4.6 Foreign ownership: treatment or selection?

Foreign firms could import more effectively either because:

- foreign acquisition *causes* better use of imports, or
- foreign investors buy firms that were already better placed to benefit from imports.

The paper uses an event study around foreign acquisition. The productivity premium of importers relative to nonimporters widens after acquisition and peaks at about **4 percentage points** around year 2.

Interpretation. The evidence is suggestive rather than definitive, but it points toward at least part of the foreign-import premium being causal.

4.7 What the paper can and cannot distinguish

- The paper can distinguish quality from imperfect substitution because it uses both productivity gains and import shares.
- It can quantify how much of aggregate productivity growth is attributable to imported inputs.
- It can show that foreign firms both use imports more effectively and face lower fixed costs.

But the paper is more limited in several respects:

- it relies heavily on the structural model in the absence of a clean external instrument;
- counterfactuals are partial equilibrium, holding input prices, capital, and labor fixed;
- the foreign-ownership event study is only suggestive about causality;
- imported capital goods are not modeled directly.

Interpretation. The paper is strongest on mechanism and quantitative decomposition within its structural framework, and weaker on fully nonparametric causal identification.

4.8 Relation to the literature

The paper links three strands of work:

- reduced-form evidence that imported inputs raise productivity (e.g., Indonesia, Chile, India);
- variety-based and quality-based trade theories;
- firm-dynamics models with heterogeneous import participation and fixed costs.

Its distinctive contribution is to estimate, in one structural model:

- the productivity effect of imported inputs,
- the quality versus complementarity decomposition,
- ownership heterogeneity,
- fixed import costs,
- and aggregate counterfactuals.

5 Tables and figures

5.1 Table 1: importers are larger, richer, and more internationally engaged

Read:

- Importers have far higher employment, sales, and sales per worker than nonimporters.
- Importers are much more likely to export and to have been foreign owned.
- Among importers, the average number of imported products is about 12.2, compared with zero for nonimporters.
- The import share in materials is 27% for importers.

Economic message. The raw data already show that importing is concentrated among larger, more internationally connected firms, which is why the paper needs a structural strategy rather than a simple importer/nonimporter comparison.

TABLE 1—DESCRIPTIVE STATISTICS

	Full sample	Nonimporters	Importers
Employment	48.85	17.42	98.87
Sales (thousands US$)	2,870	420	6,770
Capital per worker (thousands US$)	13.75	11.22	17.76
Sales per worker (thousands US$)	51.32	36.99	74.11
Material share in output	0.82	0.79	0.87
Exporter indicator	0.35	0.15	0.68
Export share in output	0.10	0.04	0.20
Importer indicator	0.39	0.00	1.00
Import share in materials	0.10	0.00	0.27
Number of imported products (HS4)	4.71	0.00	12.22
Foreign owned	0.19	0.09	0.34
State owned	0.03	0.02	0.04
Observations	127,472	78,273	49,199
Firms	26,593	20,921	13,341

Notes: Table entries are means unless otherwise noted. Column 1 is based on the full sample defined in Section IIA.

TABLE 2—IMPORT DYNAMICS

	Import growth		Contributions to import growth						
Year	Total imports (US$ millions)	Import growth (percent)	Intensive margin (pp)	New firms (pp)	New importers (pp)	New products (pp)	Stopping firms (pp)	Stopping importers (pp)	Dropped products (pp)
1992	1,968								
1993	2,358	19.8	16.2	9.6	5.1	10.6	−12.5	−2.6	−6.5
1994	3,182	35.0	28.2	3.0	5.6	8.1	−3.2	−1.3	−5.5
1995	3,885	22.1	18.6	3.3	1.9	5.6	−1.3	−1.6	−4.3
1996	5,194	33.7	12.0	2.1	4.6	24.2	−3.2	−1.0	−4.9
1997	6,668	28.4	26.1	2.0	1.2	3.9	−1.8	−0.3	−2.8
1998	8,875	33.1	32.1	0.7	0.6	3.0	−0.5	−0.3	−2.4
1999	10,799	21.7	20.4	1.4	0.4	2.2	−0.8	−0.3	−1.7
2000	13,421	24.3	21.7	8.0	0.6	1.8	−6.5	−0.1	−1.3
2001	16,413	22.3	21.3	0.2	1.6	2.8	−2.0	−0.5	−0.9
2002	15,425	−6.0	−3.6	0.4	0.4	1.2	−3.1	−0.4	−0.9
2003	14,521	−5.9	−4.5	0.4	1.0	1.1	−1.3	−0.1	−2.5
Average		20.8	17.1	2.8	2.1	5.9	−3.3	−0.8	−3.1
1992–2003		637.8	111.4	513.4	11.5	53.1	−38.2	−0.8	−12.5

Notes: Total imports are in 1998 US$. The contributions to import growth columns measure the percentage point (pp) increase in imports attributable to different mechanisms and sum to the import growth column. The intensive margin measures (net) growth in imports of products that the firm also imported the previous period (the previous year, and in the last row 1992). New firms are firms that did not exist in the previous period, new importers are firms that existed but did not import in the previous period, and new products are newly imported products of existing importers. Stopping firms, stopping importers, and dropped products are defined analogously.

5.2 Table 2: the extensive margin is quantitatively important

Read:

- Average annual import growth is 20.8%.
- The intensive margin contributes 17.1 percentage points.
- Positive extensive margins contribute 10.8 percentage points and negative extensive margins subtract 7.1 percentage points.
- The addition of new imported products contributes 5.9 percentage points on average.
- Over the full decade, much of import growth comes from new firms, reflecting major restructuring.

Economic message. Import growth is not mainly about firms buying more of the same foreign goods. It is also about firms entering importing and, crucially, about firms expanding the set of imported varieties.

5.3 Figure 1 and Table 3: the baseline import gain is large

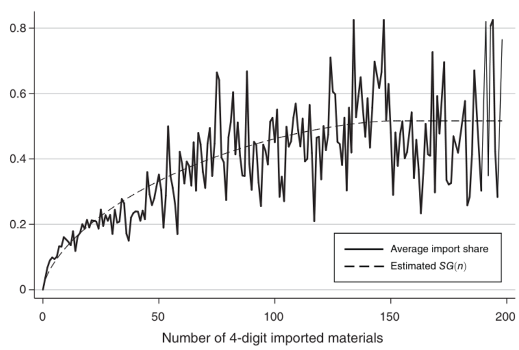

FIGURE 1. IMPORT SHARE AS A FUNCTION OF THE NUMBER OF IMPORTED PRODUCTS

TABLE 3—BASELINE ESTIMATES

Dependent variable: log sales	Baseline (1)	Conditioning on exporter (2)	Conditioning on past imports (3)	Conditioning on lagged $G(n)$ (4)
Capital (α^*)	0.041	0.041	0.040	0.038
	(0.003)	(0.003)	(0.003)	(0.004)
Labor (β^*)	0.198	0.197	0.201	0.215
	(0.008)	(0.008)	(0.008)	(0.015)
Materials (γ^*)	0.752	0.752	0.752	0.752
	(0.012)	(0.012)	(0.012)	(0.012)
Per-product import gain (a)	0.327	0.263	0.194	0.164
	(0.063)	(0.058)	(0.053)	(0.055)
Import share (S)	0.626	0.626	0.626	0.626
	(0.055)	(0.055)	(0.055)	(0.055)
Efficiency of imports (A)	1.186	1.147	1.107	1.089
	(0.076)	(0.061)	(0.047)	(0.042)
Elasticity of substitution (θ)	4.006	4.742	6.053	7.002
	[3.05; 6.07]	[3.52; 8.00]	[4.14; 12.92]	[4.45; 26.13]
Curvature of $G(n)$ (λ)	0.650	0.650	0.650	0.650
	(0.055)	(0.055)	(0.055)	(0.055)
Foreign owned	0.054	0.067	0.067	0.073
	(0.013)	(0.010)	(0.010)	(0.126)
Exporter		0.046	0.046	0.052
		(0.005)	(0.005)	(0.006)
Previous importer			−0.002	
			(0.006)	
Lagged per-product import gain				0.120
				(0.045)
Industry sales (log)	0.066	0.066	0.031	0.076
	(0.013)	(0.013)	(0.012)	(0.020)
Local demand growth	0.009	0.010	0.012	0.013
	(0.006)	(0.006)	(0.005)	(0.007)
p-value of test for $a = 0$	0.004	0.004	0.004	0.016
p-value of test for $A = 1$	0.004	0.004	0.004	0.004
Observations	127,472	127,472	127,472	127,472

Notes: All specifications use the structural estimation procedure of Section IV. Bootstrapped standard errors clustered by firm are in parentheses. For the elasticity of substitution (θ) we report a 95 percent confidence interval computed the same way in brackets.

example, $\alpha^* = \alpha(\eta - 1)/\eta$. From these, the quantity production function parameters can be recovered using our estimate of the demand elasticity η. Because the

Read:

- Figure 1 shows that the fitted $SG(n)$ function tracks the average import share reasonably well as the number of imported products rises.
- Table 3 estimates a baseline per-product import gain of $a = 0.327$.
- The implied price-adjusted quality advantage is $A = 1.186$.
- The estimated elasticity of substitution is $\theta \approx 4.0$, implying meaningful complementarity.
- Importing all tradable varieties raises revenue productivity by about 22%.

Economic message. The structural model fits the import-share data and delivers a large productivity gain from imported inputs, not merely a statistically significant one.

5.4 Figure 2 and Table 4: foreign firms get more out of imports

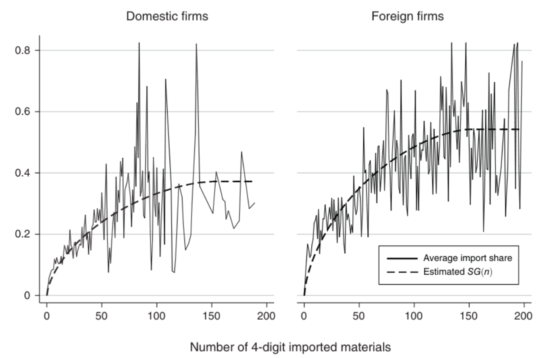

FIGURE 2. IMPORT SHARE FOR DOMESTIC AND FOREIGN FIRMS

TABLE 4—THE GAINS FROM IMPORTING FOR FOREIGN AND DOMESTIC FIRMS

	Baseline (1)		Conditioning on exporter (2)		Conditioning on past imports (3)		Conditioning on lagged $G(n)$ (4)	
Dep. variable: log sales	Domestic	Foreign	Domestic	Foreign	Domestic	Foreign	Domestic	Foreign
Capital (α^*)	0.041 (0.003)		0.041 (0.003)		0.040 (0.003)		0.039 (0.005)	
Labor (β^*)	0.199 (0.008)		0.196 (0.008)		0.201 (0.008)		0.214 (0.019)	
Materials (γ^*)	0.752 (0.012)		0.752 (0.012)		0.752 (0.012)		0.752 (0.012)	
Per-product import gain (a)	0.271 (0.063)	0.390 (0.068)	0.213 (0.057)	0.314 (0.063)	0.156 (0.052)	0.224 (0.059)	0.126 (0.062)	0.159 (0.064)
Import share (S)	0.490 (0.052)	0.621 (0.042)	0.486 (0.052)	0.625 (0.045)	0.490 (0.049)	0.621 (0.048)	0.515 (0.059)	0.596 (0.053)
Efficiency of imports (A)	0.984 (0.083)	1.220 (0.072)	0.982 (0.066)	1.178 (0.061)	0.991 (0.046)	1.121 (0.048)	1.010 (0.048)	1.071 (0.042)
Elasticity of substitution (θ)	3.484 [2.753; 5.004]		4.118 [3.135; 6.712]		5.323 [3.758; 10.919]		6.719 [4.083; 66.082]	
Curvature of $G(n)$ (λ)	0.650 (0.055)		0.650 (0.055)		0.650 (0.055)		0.650 (0.054)	
Lagged per-product import gain (a)							−0.087 (0.055)	0.306 (0.095)
Foreign owned	0.066 (0.015)		0.061 (0.014)		0.066 (0.012)		0.028 (0.077)	
Exporter			0.045 (0.006)		0.046 (0.005)		0.052 (0.006)	
Previous importer					−0.002 (0.006)			
Industry sales	0.066 (0.013)		0.066 (0.013)		0.032 (0.012)		0.070 (0.020)	
Local demand growth	0.010 (0.006)		0.010 (0.006)		0.012 (0.005)		0.012 (0.007)	
p-value of test for $a = 0$	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.219
p-value of test for $A = 1$	0.777	0.004	0.733	0.004	0.777	0.004	0.897	0.046
p-value of test for $a1 = a2$		0.012		0.016		0.012		0.269
Observations	127,472		127,472		127,472		127,472	

Notes: All specifications use the structural estimation procedure of Section IV. Bootstrapped standard errors clustered by firm are in parentheses. For the elasticity of substitution (θ) we report a 95 percent confidence interval computed the same way in brackets.

Read:

- Figure 2 shows that, for a given number of imported products, foreign firms have higher import shares than domestic firms.
- In Table 4, the baseline per-product import gain is 0.390 for foreign firms but 0.271 for domestic firms.
- The estimated import-efficiency parameter is $A = 1.220$ for foreign firms and about $A = 0.984$ for domestic firms.
- In the specification with lagged $G(n)$, the long-run import gain is much larger for foreign firms.

Economic message. Foreign firms do not just import more; they convert imports into productivity more efficiently, and this advantage remains after controlling for exporter status and past imports.

5.5 Figure 3 and Table 5: acquisition and time variation

Read:

- Figure 3 shows that the productivity premium of importers relative to nonimporters widens after foreign acquisition and peaks around year 2.
- The peak increase is roughly 4 percentage points.
- Table 5 shows that foreign firms have $A > 1$ in every subperiod, while domestic firms are close to $A = 1$.

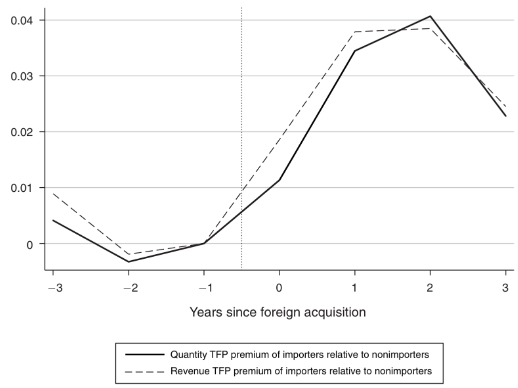

Figure 3. Productivity Premium of Importers around Foreign Acquisition

Table 5—The Gains from Importing over Time

	1992–1994	1995–1997	1998–1900	2001–2003
Domestic firms				
Per-product import gain (a)	0.301	0.287	0.244	0.314
	(0.092)	(0.069)	(0.057)	(0.069)
Import share (S)	0.513	0.497	0.443	0.529
	(0.072)	(0.058)	(0.060)	(0.065)
Efficiency of imports (A)	1.022	0.996	0.909	1.049
	(0.126)	(0.095)	(0.099)	(0.110)
Foreign firms				
Per-product import gain (a)	0.332	0.459	0.401	0.364
	(0.089)	(0.092)	(0.072)	(0.072)
Import share (S)	0.548	0.667	0.617	0.582
	(0.064)	(0.058)	(0.057)	(0.055)
Efficiency of imports (A)	1.084	1.337	1.221	1.148
	(0.118)	(0.130)	(0.102)	(0.097)
p-value of domestic $A = 1$	0.319	0.695	0.978	0.646
p-value of domestic $A =$ previous column		0.420	0.424	0.214
p-value of foreign $A = 1$	0.186	0.004	0.004	0.020
p-value of foreign $A =$ previous column		0.113	0.404	0.271
Observations		127,472		

Notes: Table reports estimates of a single regression. Different coefficients for the productivity gain from importing (a), import share (S), and efficiency of imports (A) are estimated for foreign and domestic firms in each three-year period using the structural procedure of Section IV. Other parameters are assumed to remain constant. Bootstrapped standard errors clustered by firm are in parentheses.

- The foreign advantage is especially visible in the mid-1990s and late-1990s.

Economic message. The import premium of foreign firms is not purely cross-sectional. It also seems to strengthen after foreign acquisition, which points toward a causal role for ownership-linked know-how or supplier access.

5.6 Table 7 and Figure 4: foreign firms face lower import fixed costs

Read:

- The median fixed cost of importing the *first* product is much lower for firms that are already importers than for nonimporters, consistent with selection into importing.
- The median fixed cost of importing the first product is \$35,938 for domestic nonimporters versus \$14,890 for foreign nonimporters.
- The figure shows that the estimated fixed-cost schedule rises with the number of imported products but is below the domestic schedule for foreign firms.
- Across all importers, fixed costs are about 2.2% of production costs, while the cost increase from producing without imports is about 19% of production costs.

Economic message. Fixed costs matter for the import decision, but they are small relative to the cost savings created by imported inputs. Foreign firms face systematically lower fixed costs, which helps explain why they import more varieties.

Table 7—Fixed Costs

	Domestic	Foreign
Cost of importing first product ($)		
Nonimporter	35,938	14,890
Importer	1,097	556
Cost of importing next product ($)		
	4,165	6,429

Note: Table reports the estimated median fixed costs of importing a good in 1998 US$, separately for domestic and foreign firms.

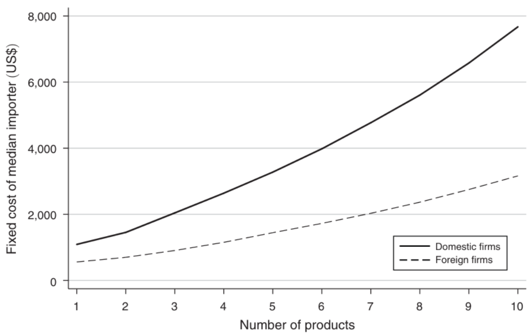

Figure 4. Fixed Cost Schedule of Domestic and Foreign Firms

Table 8—Productivity Growth in the Hungarian Manufacturing Sector 1993–2002

Growth in aggregate productivity (percent)	21.1
Coming from	
Intensive margin of import	0.7
Extensive margin of imports	4.0
Increased import by foreign firms	1.2
Direct effect of foreign firms	2.2
Other	13.0

Note: Total productivity growth between the periods 1992–1994 and 2001–2003, interpreted as growth from 1993 to 2002, decomposed into the contributions of five different margins.

the *n*th product. But because of the lower costs and the higher gain *A*, they choose a higher *n*. Hence, consistent with the bottom panel in Table 7, firms that have been foreign owned face a higher fixed cost for the next product they could be importing.

It is helpful to understand the raw fact in the data that drives these findings. The fixed costs are estimated from the comovement between the model-implied

5.7 Table 8: imports explain an important part of aggregate productivity growth

Read:

- Aggregate manufacturing productivity grows by 21.1% between 1993 and 2002.
- Import-related channels account for 5.9 percentage points.
- The largest import-related component is the extensive margin of imports, contributing 4.0 percentage points.
- Increased import use by foreign firms contributes another 1.2 percentage points.

Economic message. The firm-level import mechanism is not a small micro effect. It aggregates into a quantitatively important component of economy-wide productivity growth during Hungary's transition years.

Table 8—Productivity Growth in the Hungarian Manufacturing Sector 1993–2002

Growth in aggregate productivity (percent)	21.1
Coming from	
Intensive margin of import	0.7
Extensive margin of imports	4.0
Increased import by foreign firms	1.2
Direct effect of foreign firms	2.2
Other	13.0

Note: Total productivity growth between the periods 1992–1994 and 2001–2003, interpreted as growth from 1993 to 2002, decomposed into the contributions of five different margins.

5.8 Table 9 and Figure 5: tariff cuts interact with foreign presence and fixed costs

TABLE 9—COUNTERFACTUAL EXPERIMENTS

Tariff reduction (percent)	No firms foreign (percent)	Baseline (percent)	All firms foreign (percent)
Panel A			
40 to 30	0.8	1.3	1.6
10 to 0	1.6	2.5	2.9
	High fixed cost (percent)	Baseline (percent)	Low fixed cost (percent)
Panel B			
40 to 30	1.2	1.3	1.5
10 to 0	2.2	2.5	2.7

Notes: Table reports changes in aggregate TFP in our simulated economy in response to a 10-percentage-point tariff reduction under various scenarios. High fixed costs are three times the baseline and low fixed costs are one-third of the baseline for each firm in the simulated economy.

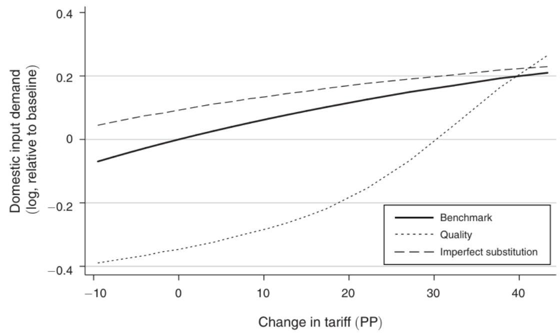

FIGURE 5. EFFECT OF TARIFF CHANGES ON DOMESTIC INPUT DEMAND

Read:

- In the baseline economy, a tariff cut from 40% to 30% raises aggregate productivity by 1.3%, while a cut from 10% to 0% raises it by 2.5%.
- Tariff cuts have larger effects when all firms are foreign than when no firms are foreign.
- Tariff cuts also have larger effects when fixed costs of importing are low.
- Figure 5 shows that domestic input demand is much more sensitive to tariffs under pure quality differences than under pure imperfect substitution.

Economic message. Trade policy is complementary to both FDI liberalization and reductions in fixed trade costs. Moreover, the redistributive consequences for domestic suppliers depend strongly on whether imports are mainly about better quality or about complementarity.

6 Short summary

- The paper estimates a structural model in which imported inputs raise productivity through quality and complementarity channels.
- Using Hungarian firm-level customs and balance-sheet data, it shows that importing all tradable varieties raises firm productivity by about one-fifth.
- Roughly one-half of the gain comes from imperfect substitution between foreign and domestic inputs.
- Foreign-owned firms both use imports more effectively and face lower fixed import costs.
- Extensive-margin import growth is quantitatively central: firms adding imported varieties matters more than simply importing more of the same goods.
- Import-related mechanisms explain more than one-quarter of Hungarian manufacturing productivity growth from 1993 to 2002.

- Tariff cuts have larger effects when fixed import costs are low and when foreign ownership is more prevalent.
- The paper's core message is that imported inputs are a major source of micro and macro productivity growth, and that their effect depends crucially on how firms organize sourcing across domestic and foreign suppliers.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that imported inputs have a large positive effect on firm productivity, that these gains are not driven solely by better foreign quality, and that foreign-owned firms benefit even more because they both use imports more effectively and face lower fixed costs of importing.

7.2 Economic intuition

Imported inputs matter along two margins. First, they may embody better or cheaper effective quality. Second, they enlarge the set of inputs available to the firm, and domestic and foreign inputs are not close enough substitutes for this diversification channel to be negligible. Because firms must also pay fixed costs to import each product, only firms with enough productivity or low enough fixed costs expand into many foreign varieties.

7.3 Takeaway

The main lesson is that trade in intermediates is a productivity technology. It shifts resources toward more productive firms, raises firm-level efficiency directly, and amplifies aggregate gains when it is combined with lower import barriers and stronger foreign participation.